# Powers of Ten and Place-Value Relationships

Grades 4–5 Number and Operations in Base Ten

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*Digit-value relationships, powers of ten in exponent form, shifting digits by 10, 100 and 1000, and rounding to a place to estimate*

**The one idea behind every problem.** Each place in our number system is worth **ten times** the place to its right, and **one tenth** of the place to its left. Everything below is that single fact used four ways.

| **hundred thousands** | **ten thousands** | **thousands** | **hundreds** | **tens** | **ones** | **tenths** | **hundredths** |
|---|---|---|---|---|---|---|---|
| $10^5$ | $10^4$ | $10^3$ | $10^2$ | $10^1$ | $10^0$ | $\frac{1}{10}$ | $\frac{1}{100}$ |
| | | | | | | | |

*Reference chart. No values are filled in — use the numbers given in each problem.*

**Directions.**

- Show the place-value thinking, not just the answer — name the places you compared or the number of places the digits moved.
- A power of ten tells you *how many places*: $10^2$ means two places, $10^3$ means three places.
- Write your final answer on the answer line.

**1.** The number 4400 is written with two 4s.

The 4 in the thousands place is worth 4000. The 4 in the hundreds place is worth 400.

How many times as much is the thousands 4 worth as the hundreds 4? Write the division you used, then give the answer.

Answer: ________________

**2.** An exponent on 10 counts how many times 10 is used as a factor.

Write $10^3$ as a repeated multiplication, then give its value in standard form.

Answer: ________________

**3.** Find $47 \times 100$.

Explain the move in place-value words: each digit of 47 shifts how many places, and in which direction? Fill in a copy of the reference chart if it helps.

Answer: ________________

**4.** The number 0.88 is written with two 8s.

The 8 in the tenths place is worth 0.8. The 8 in the hundredths place is worth 0.08.

The hundredths 8 is worth what fraction of the tenths 8? Write the division $0.08 \div 0.8$ as a decimal answer, and name the fraction it equals in words.

Answer: ________________

**5.** Estimate $4780 + 3260$ by rounding each number to the nearest thousand — that is, to the nearest $10^3$.

Write both rounded numbers, then the estimated sum.

Answer: ________________

**6.** Every whole number that ends in zeros can be written as a digit times a power of ten.

**(a)** Fill in the exponent: $60000 = 6 \times 10^{\square}$

**(b)** Multiply your $6 \times 10^{\square}$ back out to check it, and write that product on the answer line.

Answer: ________________

**7.** Find $3.6 \times 1000$.

The digits of 3.6 shift three places to the left. Show where the 3 and the 6 land, and what fills the places behind them.

Answer: ________________

**8.** The number 55500 is written with three 5s.

The 5 in the ten-thousands place is worth 50000. The 5 in the hundreds place is worth 500.

How many times as much is the ten-thousands 5 worth as the hundreds 5? Careful: the two places are not next to each other — count them.

Answer: ________________

**9.** Estimate $6238 - 2905$ by rounding each number to the nearest hundred — that is, to the nearest $10^2$.

Write both rounded numbers, then the estimated difference. Look closely at the tens place of 2905 before you round it.

Answer: ________________

**10.** Find $4500 \div 100$.

Dividing by a power of ten shifts the digits the opposite way from multiplying. Say how many places the digits move and in which direction, then give the quotient.

Answer: ________________

**11.** A number is written in expanded form using powers of ten:

$$3 \times 10^4 \; + \; 7 \times 10^2 \; + \; 5$$

**(a)** What is the value of each term?

**(b)** Write the number in standard form. Two places have no term written for them — decide what digit belongs in each of those places and why.

Answer: ________________

**12.** A warehouse counts 8120 stickers and packs them into boxes of 39.

Estimate $8120 \div 39$ by first rounding the divisor to the nearest ten, so that the division becomes a shift by a power of ten.

**(a)** What does 39 round to? **(b)** Estimate the quotient. **(c)** Is your estimate larger or smaller than the exact quotient? Explain in one sentence why rounding the divisor *up* does that.

Answer: ________________